Platform v2 Comparison Matrix

Local ACI automation project versus colleague repository

Local: C:\MCP\INFRA-AUTOMATION-PLATFORM

Colleague: github.com/panosnaz/infra-automation-platform, master, observed HEAD 09b02c8

Date: 8 September 2026

Scope: architecture, Terraform modules, MCP tools, generated intent, tests, and validation evidence.

Conclusion: Both projects use the same Platform v2 architecture. The local project has the newer network-service extensions developed in this session. The colleague repository has broader platform-policy coverage, especially COOP/ISIS, RBAC, and fault/syslog monitoring.

Comparison basis: remote repository inspected read-only. No remote code was fetched, merged, or copied.

1. Executive Matrix

Capability	Local Project	Colleague Repository	Result
Platform architecture	Platform v2: Nautobot → generator → GitLab → Terraform/Ansible/verification	Same Platform v2 architecture	**SAME**
Nautobot source of truth	Yes	Yes	**SAME**
Generated YAML	NetAsCode-style YAML	NetAsCode-style YAML	**SAME**
GitLab governance	Validation, OPA, approval, plan/apply, verification, capture	Same model	**SAME**
Physical/protocol L3Out	Implemented and plan-validated	Not found in inspected remote excerpts	**LOCAL LEAD**
VRF route leaking	Implemented and plan-validated	Not found in inspected remote excerpts	**LOCAL LEAD**
L4-L7 / PBR / Service Graph	Two-arm and one-arm logical support	Not found in inspected remote excerpts	**LOCAL LEAD**
VMM Domain	Implemented and plan-validated	Implemented with refined binding API	**BOTH / REMOTE API REFINEMENT**
COOP / ISIS / Pod Policy Groups	Not present	Implemented	**REMOTE LEAD**
RBAC / Security Domains / Local Users	Not present	Implemented	**REMOTE LEAD**
Fault / Syslog Monitoring	Not present	Implemented in Phase G	**REMOTE LEAD**
EVPN domain	Implemented and live-verified	Implemented and live-verified	**SAME**

2. Terraform Resource Matrix

Terraform area	Local resources	Remote status from inspection	Comparison
Core ACI	`aci_tenant`, `aci_vrf`, `aci_bridge_domain`, `aci_subnet`	Same resource family	Equivalent
EPG and contracts	`aci_application_profile`, `aci_application_epg`, filters, contracts, subjects, relations	Same Phase A family	Equivalent baseline
Logical L3Out	`aci_l3_outside`, external EPG, external subnet	Logical L3Out baseline	Equivalent baseline
VLAN/access policy	VLAN pool, ranges, physical domain, AAEP, IPG	Implemented/refined in remote	Compare schemas
IPG policies	CDP, LLDP, link-level, spanning-tree, port-security	Access policy work present	Compare provider relations
Leaf/static path	Leaf profile, access selector, EPG static path	Access policy work present	Compare path semantics
BGP L3Out	Node/profile/path/SVI/BGP peer resources	Not found in inspected excerpts	Local extension
OSPF L3Out	OSPF profile and OSPF policy resources	Not found in inspected excerpts	Local extension
VRF leaking	`aci_vrf_leak_epg_bd_subnet`	Not found in inspected excerpts	Local extension
L4-L7/PBR	Device, logical interface, graph, contexts, redirect policies/destinations	Not found in inspected excerpts	Local extension
VMM	Domain, credential, controller, VLAN relation	Implemented/refined	Compare binding behavior
COOP/ISIS	None	`aci_coop_policy`, `aci_isis_domain_policy`, Pod Policy Groups	Remote extension
RBAC	None	AAA domain, local user, user-domain, roles	Remote extension
Fault/Syslog	None	Fault lifecycle and syslog singleton resources	Remote extension

3. MCP Tool Matrix

Tool group	Local project	Colleague repository	Difference
Core objects	`create_tenant`, `create_vrf`, `create_bridge_domain`, `create_epg`	Same core tools	Same
Contracts/L3Out	`create_contract`, `create_l3out`, `create_vrf_route_leak`	Core Contract/L3Out tools; no route-leak tool observed	Local adds route leaking
Access objects	VLAN pool, physical domain, AAEP, IPG, leaf profile, selector, static path	VLAN pool, physical domain, AEP, IPG and refined access tools	Similar, different API shape
EPG domain binding	Separate physical/VMM binding tools	One `bind_epg_domain` with `domain_type`	Remote consolidated API
BGP/OSPF L3Out	Full requested tool family, including peers and OSPF policy	Not observed in inspected remote excerpts	Local lead
L4-L7/PBR	Two-arm, one-arm, device, graph, policy, contract tools	Not observed in inspected remote excerpts	Local lead
VMM	`create_vmm_domain`	`create_vmm_domain`	Remote binding refinement
RBAC	None	`create_security_domain`, `create_local_user`	Remote lead
Status	`show_status`	`show_status`	Same

4. Generator and YAML Matrix

Intent domain	Local YAML model	Remote YAML model	Decision
Location fabric policy	VLAN/access objects, VMM, IPG policy values	Also COOP/ISIS, Pod Policy Groups, monitoring policies	Merge additively
Tenant policy	VRFs, BDs, EPGs, contracts, L3Outs, route leaks, L4-L7, OSPF policies	Same baseline plus later policy phases	Preserve both extensions
EPG binding schema	Separate physical/VMM fields plus static paths	Domain-aware `aci_epg_domains` pattern observed	Schema hotspot
L3Out protocol schema	Node profiles, interfaces, BGP peers, OSPF entries	Not observed in inspected remote excerpts	Retain local model
Generated artifact	Generated NetAsCode YAML under `platform/netascode/aci`	Same path and concept	Shared contract

5. Unit-Test Matrix

Test area	Local coverage	Remote coverage	Comparison
Transformer	`tests/unit/test_transformer.py`; 25 passing in latest focused run	Same file plus COOP/ISIS, Pod Policy, AAA, fault/syslog tests	Remote broader platform coverage
MCP schemas	`test_schemas_aci.py`	Same file plus RBAC/access schemas	Remote broader administration coverage
MCP tools	`test_tools_aci.py` covers local catalogue	Tests access, VMM binding, RBAC, local users	Different feature emphasis
Registry/signature	`test_registry.py`; 39 focused MCP tests passing with schema/tool suite	Same registry quality gate	Same control
EVPN	Transformer, MCP, and pyATS-equivalent tests	Transformer, MCP, integration, and verification tests	Both cover EVPN
Terraform plan evidence	Access/L3Out: 22; OSPF: 23; IPG: 27 resources added in temporary-state plans	Broader historical phase evidence	Different validated scopes

6. Strengths and Gaps Matrix

Solution	Strengths	Gaps	Recommended role
Local checkout	Network-service depth: physical/protocol L3Out, route leaks, L4-L7/PBR, one-arm ADC, OSPF/IPG policies, lab templates.	Missing remote COOP/ISIS, RBAC, and fault/syslog phases; new physical scope is plan-validated but not apply-verified.	Use for developing and validating the supplied network-service scenarios.
Colleague repository	Broader platform administration and monitoring coverage; mature later ADR phases; established history.	Local L3Out protocol, route-leak, and L4-L7 extensions were not evident in inspected excerpts.	Use as reference for missing platform-wide ACI policy phases.

7. Reconciliation Matrix

Step	Action	Reason	Acceptance check
1	Compare remote Phase E/F/G Terraform sections with local `main.tf`.	Identify missing platform-policy resources.	Resource and provider schemas reviewed side by side.
2	Compare transformer custom-field shapes.	Avoid breaking generated YAML.	Old fixtures and new fixtures both pass.
3	Reconcile MCP names and request schemas.	Remote `bind_epg_domain` differs from local split tools.	Catalogue and registry tests pass.
4	Add unit tests before merging features.	Protect both existing and new domains.	Transformer, schema, tool, registry suites pass.
5	Run temporary-state APIC plans.	Catch provider enum/relation errors safely.	Plan has no errors; no apply performed.
6	Apply only after explicit approval.	Prevent competing Terraform ownership.	One state, one owner, reviewed plan.

8. Recommendation

Use the local project as the working branch for the supplied physical L3Out, BGP/OSPF, VRF leaking, L4-L7/PBR, and OSPF-policy scenarios. Use the colleague repository as the reference branch for COOP/ISIS, RBAC/security domains/local users, and fault/syslog monitoring. Reconcile by capability and schema, not by copying the remote `main.tf` or MCP files wholesale.

> **Operational rule:** one ACI object must have one authoritative Terraform state and one desired-state path. Never allow both repository roots to manage the same tenant, VRF, L3Out, or policy object simultaneously.

9. Evidence

Evidence type	Local reference	Remote reference
Terraform	platform/terraform/aci/main.tf, providers.tf, variables.tf	platform/terraform/aci/main.tf at remote HEAD 09b02c8
MCP	mcp-server/src/mcp_server/tools/aci.py, schemas, Nautobot client	Remote ACI tools/schemas and tool tests
Generator	platform/python/generator/transformer.py, tests/unit/test_transformer.py	Remote transformer and unit tests, including later Phase E/F/G keys
Architecture	ADR-016 through ADR-021 and Platform Status	Remote ADR/history, latest Phase G commit

Prepared 8 September 2026 from read-only inspection. No remote code was fetched, merged, or copied.